Computational Analysis of the ZWINT Gene Mutational Landscape in Human Cancers

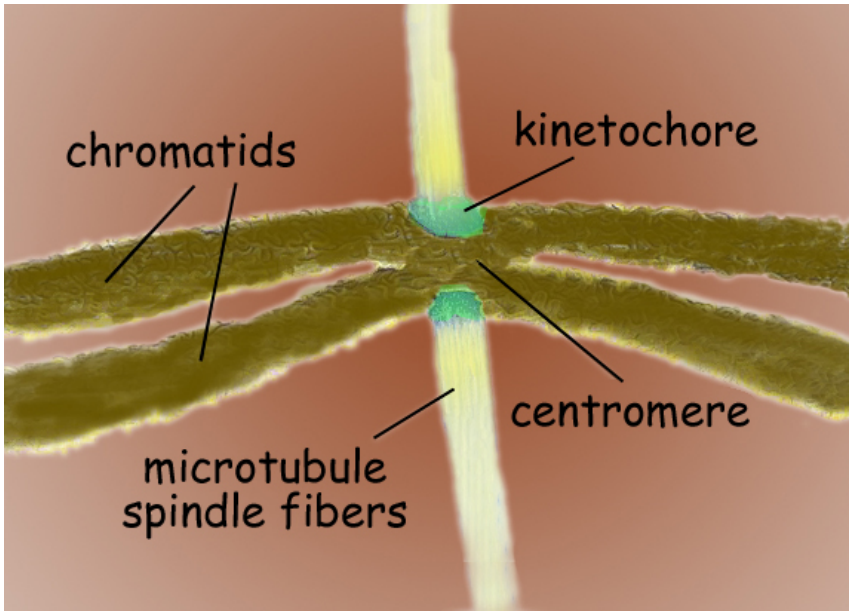
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Biological Datasets for Computational Physics

Outline

- Biological context: ZWINT and the kinetochore
- Research question
- Data and computational pipeline
- Statistical approach: χ^2 tests and Monte Carlo
- Results: tissue distribution
- Results: sequence features
- Results: positional profile
- Results: the p.R187G hotspot
- Discussion, limitations, and conclusions

ZWINT and the kinetochore



The kinetochore couples chromosomes to spindle microtubules during mitosis.

ZWINT $\rightarrow$ required for correct kinetochore–microtubule attachment.

Loss of function $\rightarrow$ disrupted attachment $\rightarrow$ chromosome missegregation $\rightarrow$ aneuploidy.

ZWINT mRNA and protein are recurrently overexpressed across several solid tumors (lung, breast, etc.).

Research question

What remains unexplored is whether this is accompanied by a distinctive signature at the **DNA** level.

Does the somatic mutational landscape of ZWINT show evidence of positive selection or is it compatible with a diffuse, passenger-like pattern?

We will study:

- tissue-specific concentration
- possible dominant mutation type
- possible spatial clustering

It is known that the majority of somatic mutations found in tumor genomes are known to be passenger rather than driver events. $\rightarrow$ Does it apply also to ZWINT?

Data and computational pipeline

Data retrieved exclusively from **COSMIC** (Catalogue of Somatic Mutations in Cancer):

- `cosmic-gm.csv`: histology and primary tissue metadata
- `cosmic-variants.csv`: variant position, CDS/AA change, mutation type

Three computational sub-tasks, each targeting one aspect of the question:

1. Group mutations by primary tissue $\rightarrow$ anatomical concentration
2. Categorize by sequence-consequence type $\rightarrow$ functional-class composition
3. Map mutation counts onto protein positions $\rightarrow$ spatial clustering

Statistical approach: χ^2 tests

Initial question:
Is one category being over-represented more than we'd expect if mutations fell randomly across all categories?

For tissue distribution and mutation type (categorical variables with a handful of classes) $\rightarrow \chi^2$ goodness-of-fit test applied against the null hypothesis of uniform distribution.

Statistical approach: Monte Carlo permutation

Initial question:
Is there any position with a remarkable amount of counts w.r.t. the others? If so, could it be "by chance" or is it "by necessity"?

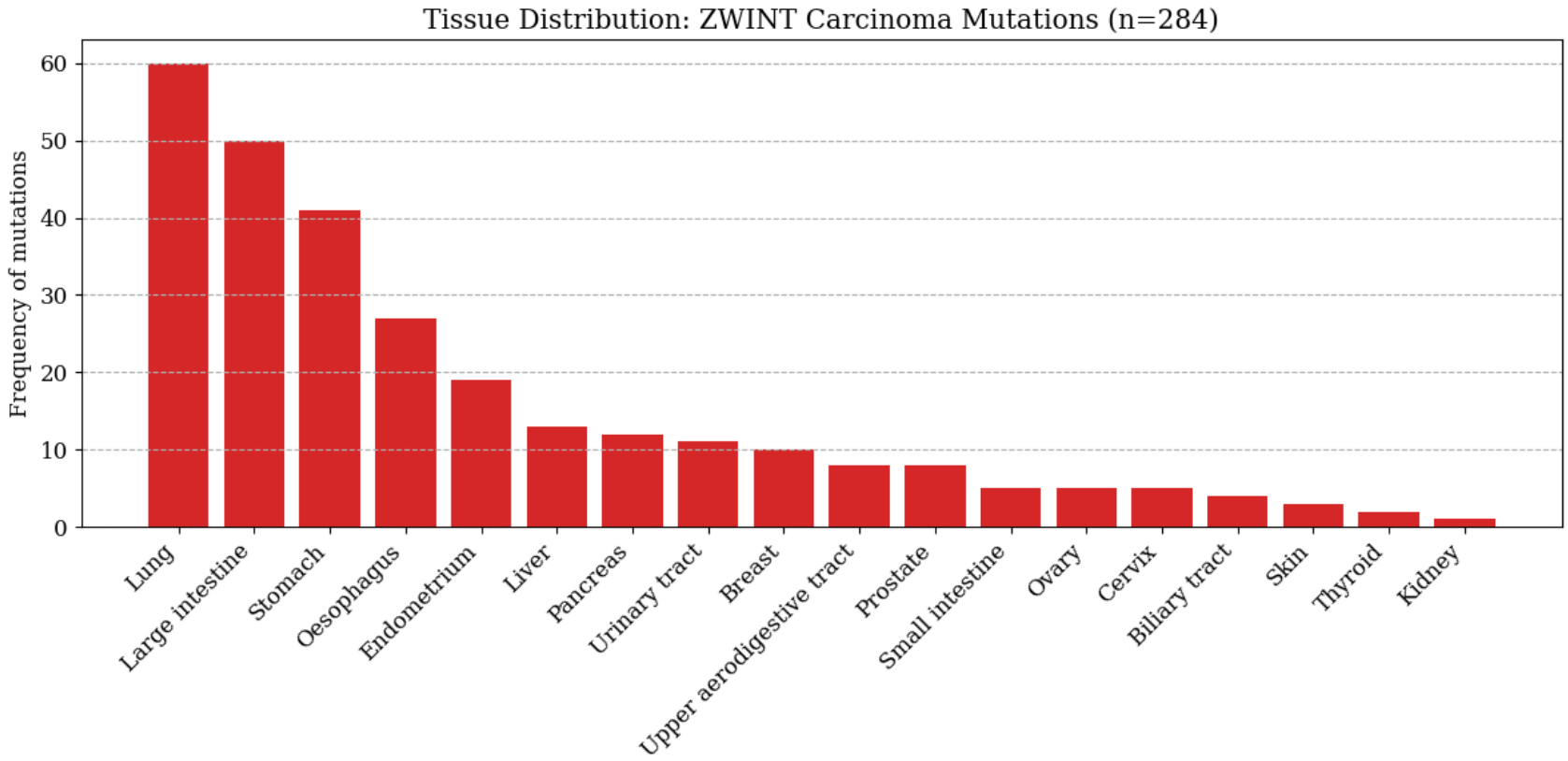
Protein position is a **continuous coordinate**, not a small set of categories → not the same approach as before

Instead, we apply a Monte-Carlo approach:

- Take the 284 observed mutations and redistribute them uniformly at random across the ∼300-residue sequence, 10^4 times. Record the maximum count reached at any single position each time.

→ Empirical null distribution for "how high a peak can get by pure chance?"

Results: tissue distribution

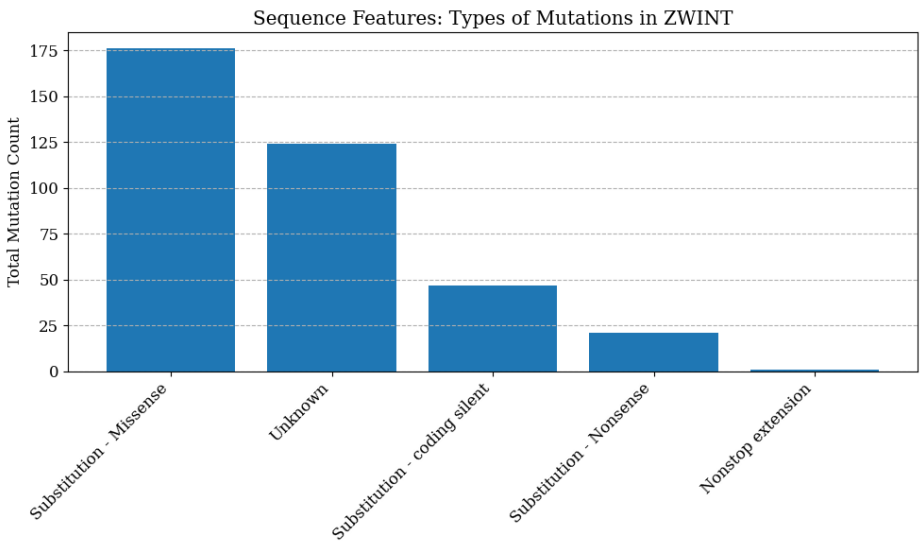


Main concentrations:**lung** (n=60), **large intestine** (n=50), **stomach** (n=41)

$\chi^2 = 326.86$, $p \approx 3 \times 10^{-59}$ $\rightarrow$ significantly non-uniform.

High-turnover, high-proliferation epithelial tissues – consistent with elevated replication-associated mutability, not a single tissue-specific driver role.

Results: sequence features

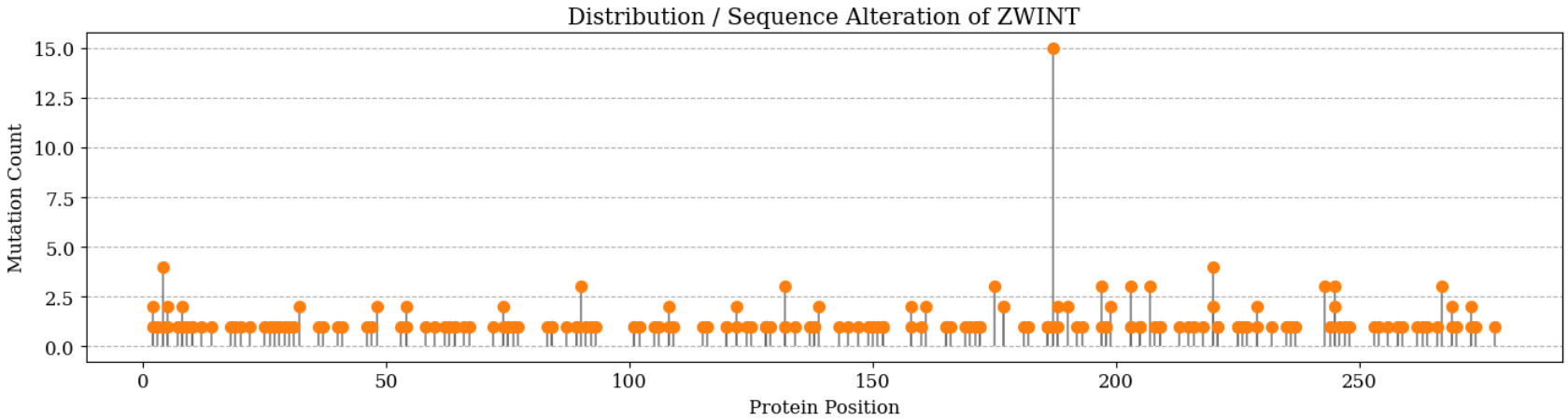


Substitution-missense dominates (n=176 $\equiv$ more than the next two categories combined)

$\chi^2 = 295.00$, $p \approx 1.3 \times 10^{-62}$.

Missense changes preserve the reading frame. Compatible with a passenger-dominated profile.

Results: positional profile



Mutation counts onto integer positions along ZWINT protein:

- Most positions carry between 1 and 4 mutations $\equiv$ Background of uniformly scattered, passenger-like variants.
- One position stands out: residue 187, with 15 mutations.

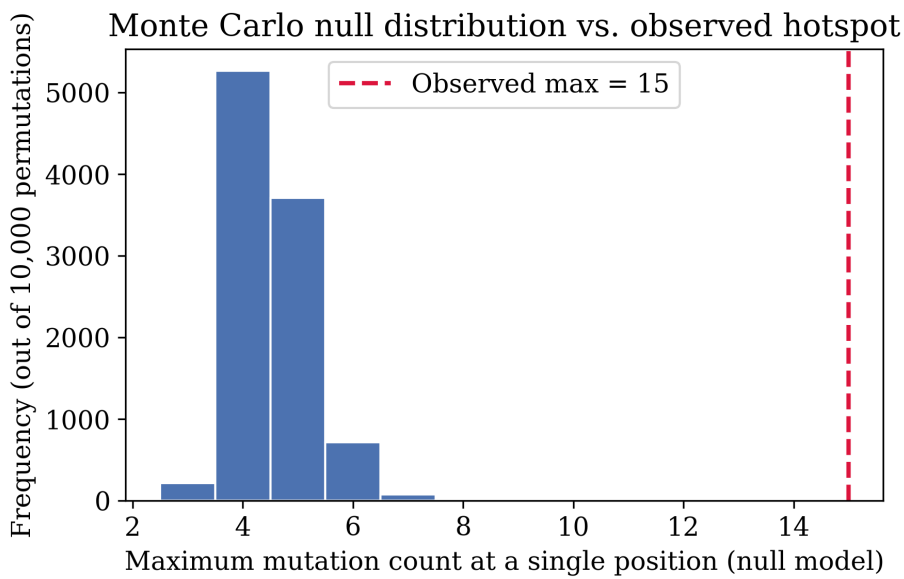
Is this within the range of chance, or not?

Results: the p.R187G hotspot

After 10^4 random resamples of the total counts: expected maximum count at any position = **4.52**.

Observed maximum (15, at residue 187) is never reached by chance: $p < 10^{-4}$.

Meets a standard criterion used in cancer genomics to flag a **candidate hotspot**.



Discussion

- Tissue-level concentration agrees with the chromosomal instability (CIN) framework [1]: genes involved in chromosome segregation, like ZWINT, can track genomic instability without acting individually as classical drivers.

- The p.R187G recurrence meets the standard criteria for a hotspot – but recurrence alone is not proof of selection. Hotspots can also arise from locally elevated background mutability rather than true selection [2]

  $\Rightarrow$ Without an independent test of local mutability, residue 187 is best reported as a **candidate hotspot**, not a confirmed driver.

[1] Carter et al.(2006) [2] Hess et al.(2019)

Limitations

- Used only COSMIC records $\rightarrow$ subject to database curation (e.g., 124 mutations classified as "Unknown" type).
- Our Monte Carlo hotspot test did not account for sequence-context effects that could locally elevate mutation rates without selection.
- This kind of hotspot detection, on its own, cannot distinguish a functionally selected driver from a locally mutable passenger site.
  $\rightarrow$ Structural or functional follow-up would be needed.

Conclusions

ZWINT's mutational landscape follows a **two-component model**:

broad passenger background (systemic chromosomal instability)
+
one statistically robust, non-random substitution.

Future work:

- Map p.R187G onto 3D structural models of ZWINT
- Apply a sequence-context-aware mutation model to test local mutability directly